

Technological progress and growth

The general Solow model

In one respect the basic Solow model presented in Chapter 3 did not perform well empirically: its long-run balanced growth path displayed zero growth in GDP per capita. This is at odds with the observed long-run growth in living standards in Western economies.

We want to develop a growth model whose long-term prediction is a balanced growth path with strictly positive growth in GDP per person. Obviously we can only trust the answers to our big questions, 'what creates prosperity in the long run?' and 'what creates transitory and long-run growth?', if these answers are grounded in a (growth) model that accords with the most basic empirical facts.

This chapter presents a growth model with the properties we are looking for: the steady state of the model exhibits balanced growth with positive growth in output per worker. We obtain this by a slight generalization of the basic Solow model. The resulting model is close to the one actually suggested by Robert M. Solow in his famous 1956 article referred to in Chapter 3.¹

The essential new feature of the model is that total factor productivity is no longer assumed to be a constant, B . Instead it will be given as an exogenous sequence, (B_t) , which may be steadily growing over time. In that case the model's steady state will display balanced growth with steady positive growth in GDP per worker.

Hence, according to the general Solow model (as we will call it) *the root of steady positive long-run growth in GDP per person is a steady exogenous technological progress*. This explanation of growth may not seem deep. However, it is not trivial that the consequence of steadily arriving technological progress should be a *balanced* growth path, and it is reassuring for the application of the model to issues of economic policy that its steady state mirrors a robust long-run growth fact. Of course, *explaining* the technological progress that creates growth in GDP per worker is a matter of great interest and it is the subject of Part 3 of this text.

Having obtained in this chapter a model that fits the basic stylized growth facts better, we are going to take the model through some more specific empirical tests. The tests will focus on the steady state prediction of the model as well as its outside steady state prediction concerning transitory growth or convergence.² Our conclusion will be that the general Solow

1. For your convenience, we give the reference to this pathbreaking article once again: Robert M. Solow, 'A Contribution to the Theory of Economic Growth', *Quarterly Journal of Economics*, 70, 1956, pp. 65–94.

2. We could have done some of these more specific tests already in connection with the basic Solow model of Chapter 3, and, in fact, Exercise 9 of Chapter 3 asked you to do one of them yourself, but we think it is better to undertake them now that the model is in accordance with the basic fact of steady and positive long-run growth.

model does quite well empirically, both with respect to its steady state and its transitory growth predictions, although there will be some aspects still to be improved upon. The general Solow model is indeed a very important growth model.

5.1 The general Solow model

With respect to the qualitative features of the underlying 'micro world', the general Solow model is identical to the basic Solow model. It has the same commodities and markets, and the markets are again assumed to be perfectly competitive. There are also the same kinds of economic agents, and their behaviour is essentially the same. In particular, a representative profit-maximizing firm has to decide on the inputs of capital and labour services, K_t^d and L_t^d , in each period t , given the real rental rate of capital, r_t , and the real wage rate, w_t . The only difference is that the production function, which tells how much output can be produced from K_t^d and L_t^d , may now change over time so that, for instance, more and more output can be obtained from the same amounts of inputs.

The production function with technological progress

The total factor productivity will now be allowed to depend on time. The notation for it will therefore be B_t , where $B_t > 0$ in all time periods t . The full time sequence, (B_t) , of total factor productivity is exogenous. Maintaining a Cobb–Douglas form, the production function in period t is:

$$Y_t = B_t K_t^\alpha L_t^{1-\alpha}, \quad 0 < \alpha < 1. \quad (1)$$

This time we insert from the beginning the feature that the inputs demanded and actually used, K_t^d and L_t^d , must equal the (inelastic) supplies, K_t and L_t , because the factor markets clear. Since one particular assumption on (B_t) could be that B_t is constant, the general Solow model, or just the Solow model, is a *generalization* of the basic Solow model.

We may alternatively write the production function as:

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}, \quad (2)$$

where $A_t \equiv B_t^{1/(1-\alpha)}$. With the Cobb–Douglas form of the production function it makes no difference whether we describe technological change by a certain time sequence, (B_t) , of the total factor productivity, or by an appropriately defined sequence, (A_t) , of the labour productivity variable (not to be confused with the average labour productivity, Y_t/L_t).

With more general production functions it may make *some* difference. Generally, technological progress that appears as an increasing variable, A_t , in a production function, $F(K_t, A_t L_t)$, is called labour-augmenting or Harrod-neutral. If it appears as an increasing variable, D_t , in $F(D_t K_t, L_t)$, it is called capital-augmenting or Solow-neutral, while, when it appears as an increasing variable, B_t , in $B_t F(K_t, L_t)$, it is called Hicks-neutral.

Given the Cobb–Douglas specification one can simply choose the formulation of technological progress that is most convenient. For our present purposes this turns out to be the form with labour-augmenting technological change as in (2).

A full description of the production possibilities includes a specification of the exogenous sequence, (A_t) . We will simply assume that the labour productivity variable, A_t , is changing at a constant rate:

$$A_{t+1} = (1 + g)A_t, \quad g > -1. \quad (3)$$

Here g is the exact (not the approximate) growth rate of A_t . It follows that if the technological level in some initial period zero is A_0 , then in period t it is $A_t = (1 + g)^t A_0$. For the approximate growth rate, $g_t^A \equiv \ln A_t - \ln A_{t-1}$, one has from (3), $g_t^A = \ln(1 + g) \approx g$ (where we use the approximation, $\ln v \approx v - 1$, as in Chapter 2).

We thus associate technology with the level of A_t and technological progress with an increase in A_t , but recall from Chapter 3 that A_t really has a residual interpretation as the influence of ‘all other factors’. A positive value of g corresponds to a steadily arriving technological progress that comes exogenously to the economy without this requiring the use of economic resources. The production function just becomes ever more efficient period by period. It is sometimes said that technological progress comes as manna from heaven. The idea of the ‘endogenous growth’ models presented in Part 3 is exactly to change the description of technological progress, so that it will be the outcome of a use of economic resources.

We will again use the definitions of output per worker (average labour productivity) $y_t \equiv Y_t/L_t$, and capital per worker (the capital intensity) $k_t \equiv K_t/L_t$. Dividing by L_t on both sides of (2) gives the ‘per capita production function’:

$$y_t = k_t^\alpha A_t^{1-\alpha}, \quad (4)$$

and then, taking logs and time differences,

$$\ln y_t - \ln y_{t-1} = \alpha(\ln k_t - \ln k_{t-1}) + (1 - \alpha)(\ln A_t - \ln A_{t-1}), \quad (5)$$

or in the usual notation for approximate growth rates:

$$g_t^y = \alpha g_t^k + (1 - \alpha)g_t^A. \quad (6)$$

These expressions reveal that an increase in output per worker can be obtained in two ways, by more capital per worker or by better technology. In fact, the (approximate) growth rate in output per worker is the weighted average of the rates of growth in capital per worker and in technology, the weights being α and $(1 - \alpha)$, respectively. There are now two potential sources of economic growth: capital accumulation and technological progress.

The complete model

Since the general Solow model is in all respects, other than the production function, identical to the basic Solow model, we can write down the full model consisting of seven equations without further delay:

THE COMPLETE GENERAL SOLOW MODEL



$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}, \quad (7)$$

$$r_t = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1}, \quad (8)$$

$$w_t = (1 - \alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha A_t, \quad (9)$$

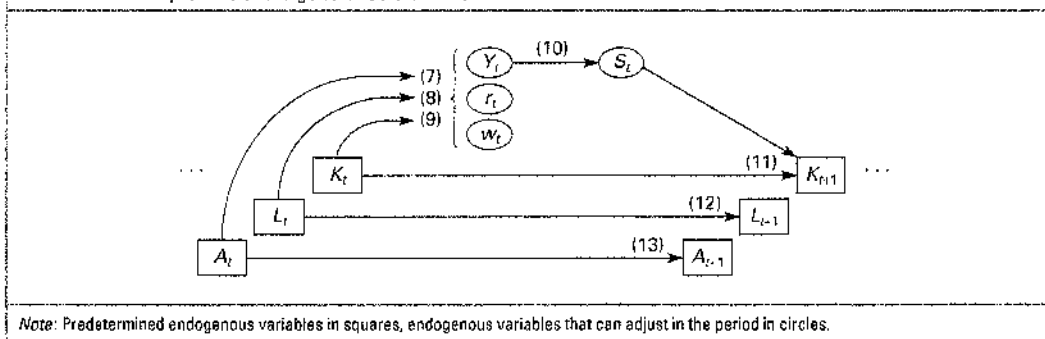
$$S_t = s Y_t, \quad (10)$$

$$K_{t+1} - K_t = S_t - \delta K_t, \quad K_0 \text{ given} \quad (11)$$

$$L_{t+1} = (1 + n)L_t, \quad L_0 \text{ given} \quad (12)$$

$$A_{t+1} = (1 + g)A_t, \quad A_0 \text{ given} \quad (13)$$

FIGURE 5.1 The dynamics of the general Solow model



The first equation is a repetition of the production function with the input supplies inserted. The next two equations give the rental rates from the marginal products of the inputs. As in the basic Solow model this follows from competitive clearing of the input markets, but now the expressions for the marginal products are slightly different, since they are derived from the new production function. Note that we still have $r_t K_t / Y_t = \alpha$ and $w_t L_t / Y_t = 1 - \alpha$, so the functional income distribution is still given by α , and there are no pure profits. Equations (10), (11) and (12) are exactly as before, so there is no change in the assumed behaviour concerning savings or 'fertility'. Of course, the fundamental capital accumulation equation, (11), has to be respected as before. The last equation is just the addition of the assumption on technological change.

For given initial values, K_0 , L_0 , A_0 , of the state variables, the model (7)–(13) determines the full dynamic evolution of the economic variables as illustrated in Fig. 5.1.

We have chosen to give a discrete time formulation of the general Solow model, but in an exercise you will be confronted with the counterpart model in continuous time.

5.2 Analysing the general Solow model

Comparing the general Solow model in the box above to the basic Solow model of Chapter 3, you will see that the two models are mathematically equivalent: in the general Solow model the product $A_t L_t$ has taken the place of L_t in the basic Solow model, and the exogenous growth factor of $A_t L_t$ is $(1 + n)(1 + g)$, rather than just $1 + n$. The analysis of the model should therefore more or less 'translate', letting $A_t L_t$ take the place of L_t from Chapter 3's analysis.

The law of motion

We analysed the basic Solow model in terms of the variables $k_t \equiv K_t / L_t$ and $y_t \equiv Y_t / L_t$, which turned out to converge to specific and constant steady state levels. A qualified guess is thus that the general Solow model can be analysed in terms of the variables:

$$\tilde{k}_t \equiv \frac{K_t}{A_t L_t} = \frac{k_t}{A_t} \quad \text{and} \quad \tilde{y}_t \equiv \frac{Y_t}{A_t L_t} = \frac{y_t}{A_t}, \quad (14)$$

that is, the technology-adjusted capital intensity (capital per *effective* worker) and the technology-adjusted average labour productivity (output per *effective* worker), respectively, and that these will converge to constant steady state values.

Note that this property would indeed be highly desirable, since if both k_t / A_t and y_t / A_t are constant in steady state, then both k_t and y_t must be increasing at the same rate as A_t , the rate g . We then get a steady state with a constant growth rate of income per worker *and* a constant capital–output ratio, k_t / y_t , in accordance with balanced growth.

Dividing on both sides of the production function (7) by $A_t L_t$ gives:

$$\bar{y}_t = \bar{k}_t^\alpha. \quad (15)$$

Now, start again from the capital accumulation equation with the savings behaviour inserted: $K_{t+1} = sY_t + (1 - \delta)K_t$. Divide on both sides by $A_{t+1}L_{t+1}$. On the left-hand side you will now have $\bar{k}_{t+1}$. On the right-hand side use $A_{t+1}L_{t+1} = (1 + n)(1 + g)A_t L_t$ to get:

$$\bar{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s \frac{Y_t}{A_t L_t} + (1-\delta) \frac{K_t}{A_t L_t} \right) = \frac{1}{(1+n)(1+g)} (s\bar{y}_t + (1-\delta)\bar{k}_t).$$

Finally insert that $\bar{y}_t = \bar{k}_t^\alpha$ to find:

$$\bar{k}_{t+1} = \frac{1}{(1+n)(1+g)} (s\bar{k}_t^\alpha + (1-\delta)\bar{k}_t). \quad (16)$$

This is the law of motion, or the *transition equation*, in the state variable $\bar{k}_t$.

From given initial values of capital, labour and technology, K_0 , L_0 and A_0 , an initial value for capital per effective worker, $\bar{k}_0 = K_0/(A_0 L_0)$, will be given. The first-order difference equation (16) will then determine $\bar{k}_1$ and then the full sequence, $(\bar{k}_t)$. An associated sequence, $(\bar{y}_t)$, follows from $\bar{y}_t = \bar{k}_t^\alpha$. From these, the sequences for the economic variables in which we are interested follow. For instance, from (8) and (9):

$$r_t = \alpha \bar{k}_t^{\alpha-1}, \quad (17)$$

$$w_t = (1-\alpha)A_t \bar{k}_t^\alpha. \quad (18)$$

Further, since income per worker is $y_t = \bar{y}_t A_t$, and $A_t = (1+g)^t A_0$ we have:

$$y_t = \bar{k}_t^\alpha (1+g)^t A_0,$$

and then consumption per worker follows as $c_t = (1-s)y_t$. Given initial values K_0 , L_0 , A_0 , the transition equation (16) and other model elements determine the full dynamic evolution of the economy.

Subtracting $\bar{k}_t$ on both sides of (16) gives the *Solow equation*:

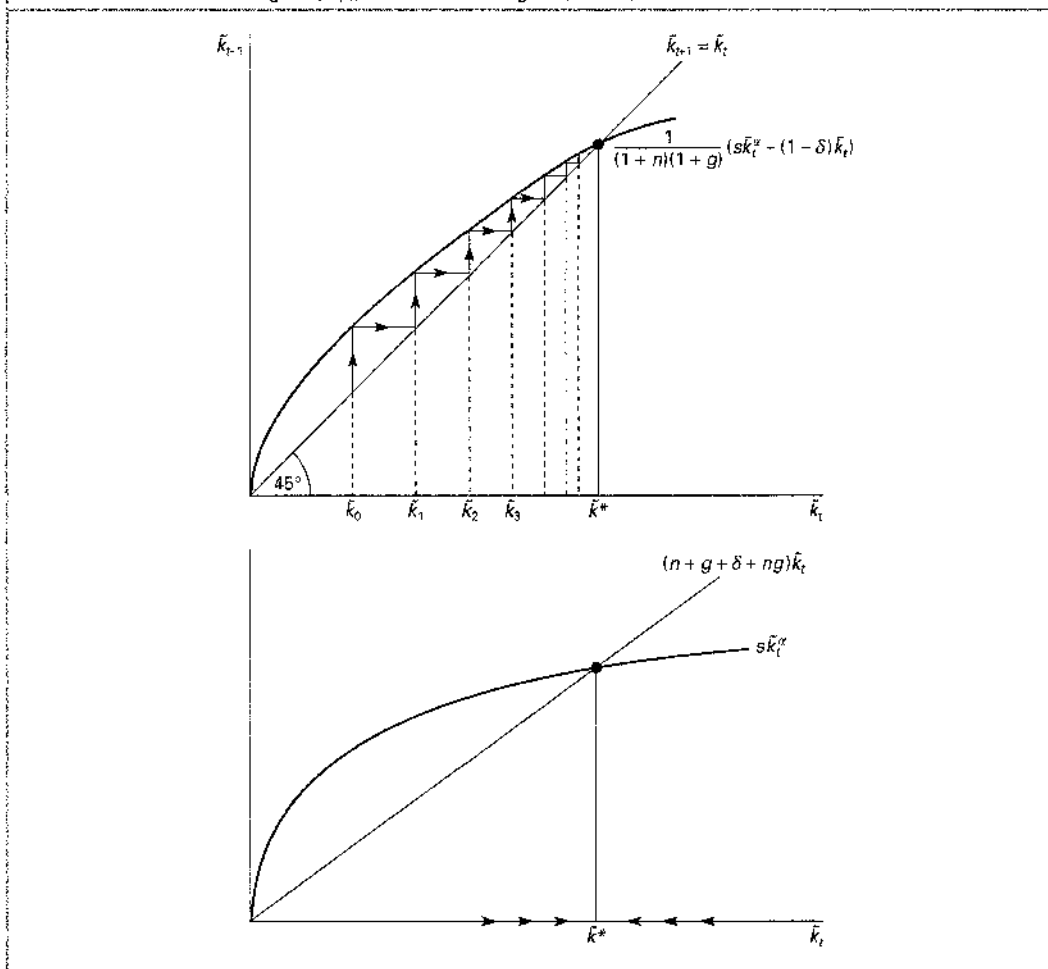
$$\bar{k}_{t+1} - \bar{k}_t = \frac{1}{(1+n)(1+g)} (s\bar{k}_t^\alpha - (n+g+\delta+ng)\bar{k}_t). \quad (19)$$

The transition equation (16) and the Solow equation (19) look much like their counterparts for the basic Solow model of Chapter 3, only now they incorporate technological change appropriately. Thus they involve $\bar{k}_t$ rather than k_t , and the expressions involve the parameter g . However, qualitatively and with respect to interpretation the laws of motion are similar to those in the basic Solow model.

Convergence to steady state

The *transition diagram* associated with (16) and the *Solow diagram* associated with (19) also look much like their counterparts for the basic Solow model, as illustrated in Fig. 5.2. In particular, the slope of the transition curve shown in the upper part of the figure decreases monotonically from infinity to $(1-\delta)/[(1+n)(1+g)]$, as $\bar{k}_t$ increases from zero to infinity (show this). The limit slope as $\bar{k}_t$ approaches infinity is less than one if $n+g+\delta+ng > 0$, which is highly plausible. Hence, under this realistic stability condition the transition curve

FIGURE 5.2 The transition diagram (top), and the Solow diagram (bottom)



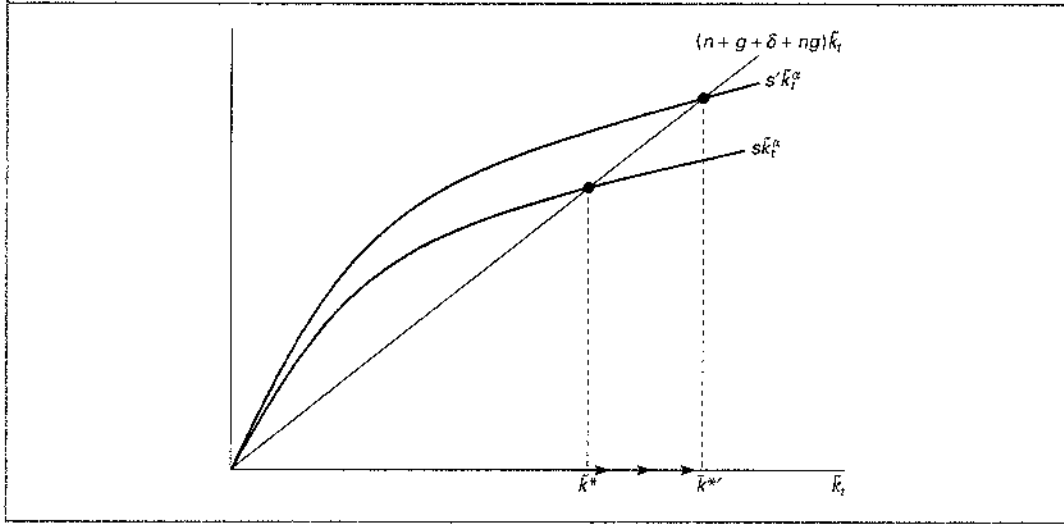
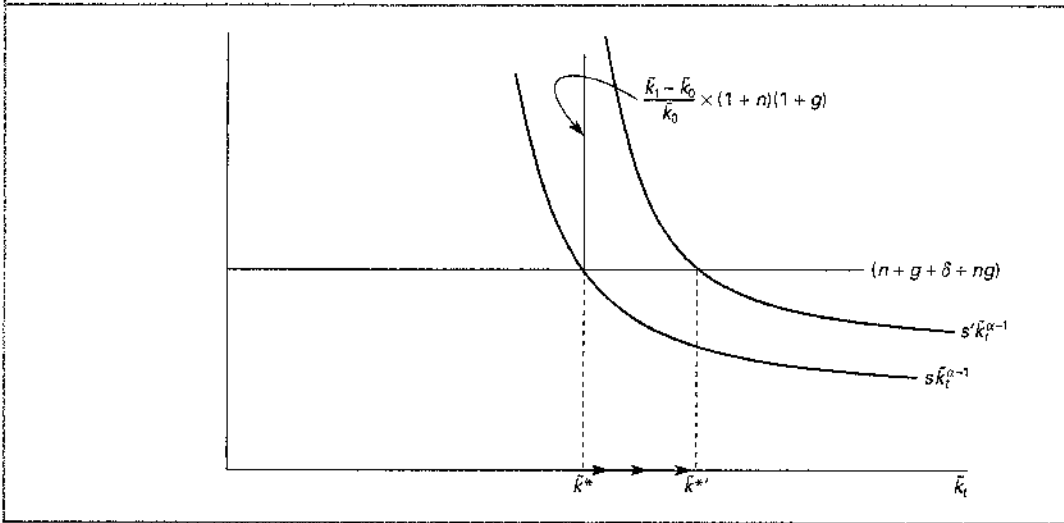
has the properties illustrated in the upper part of Fig. 5.2, and from these properties convergence to a steady state, given by the unique positive intersection between the transition curve and the 45° line, follows simply by 'staircase iteration' in the figure:

CONVERGENCE TO STEADY STATE



Given the plausible stability condition, $n + g + \delta + ng > 0$, the technology-adjusted capital intensity, $\bar{k}_t \equiv k_t/A_t$, will converge to a specific steady state value, $\bar{k}^*$, and income per effective worker, $\bar{y}_t \equiv y_t/A_t$, will converge to its steady state value, $\bar{y}^* = (\bar{k}^*)^\alpha$. Hence in the long run both capital per worker, k_t , and income per worker, y_t , converge on growing at rate g , implying that the capital–output ratio, $k_t/y_t = K_t/Y_t$, converges to a constant value in accordance with balanced growth.

The corresponding adjustments in the Solow diagram have been indicated by arrows in the lower part of Fig. 5.2. As long as the curve $s\bar{k}_t^\alpha$ lies above the ray $(n + g + \delta + ng)\bar{k}_t$, we must have $\bar{k}_{t+1} > \bar{k}_t$, so $\bar{k}_t$ must be increasing, and vice versa.

FIGURE 5.3 An increase in the savings rate in the Solow diagram**FIGURE 5.4** An increase in the savings rate in the modified Solow diagram

Comparative analysis in the Solow diagrams

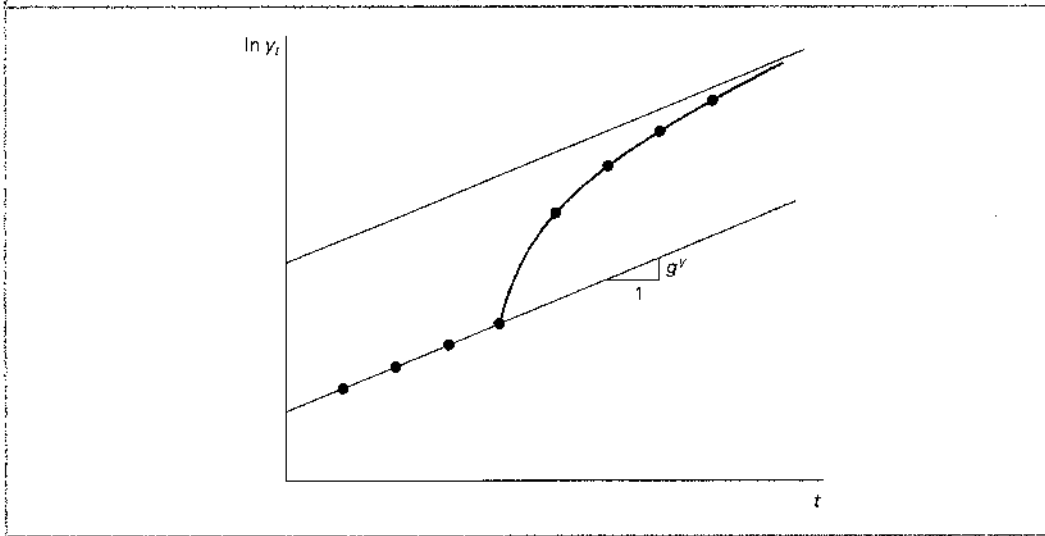
Figure 5.3 illustrates the effects of an increase in the savings rate from s to s' in the Solow diagram. The curve $s\bar{k}_t^\alpha$ shifts upwards to $s'\bar{k}_t^\alpha$, implying a new and higher steady state level, $\bar{k}^{**}$, of capital per effective worker.

In Fig. 5.4 we illustrate the shift in s in the *modified Solow diagram* associated with the *modified Solow equation*, (20), which is derived from (19) by dividing by $\bar{k}_t$ on both sides:

$$\frac{\bar{k}_{t+1} - \bar{k}_t}{\bar{k}_t} = \frac{1}{(1+n)(1+g)} (s\bar{k}_t^{\alpha-1} - (n+g+\delta+ng)) \quad (20)$$

Assume that the economy is initially in the old steady state where $\bar{k}_t$ and $\bar{y}_t$ are constant and equal to $\bar{k}^*$ and $\bar{y}^*$, respectively. In this situation capital per worker, k_t , and output per

FIGURE 5.5 Transition from the old to the new steady state



worker, y_t , both change over time at the rate g which we assume to be positive. The growth path corresponding to the old steady state is illustrated by the lower upward-sloping line in Fig. 5.5.

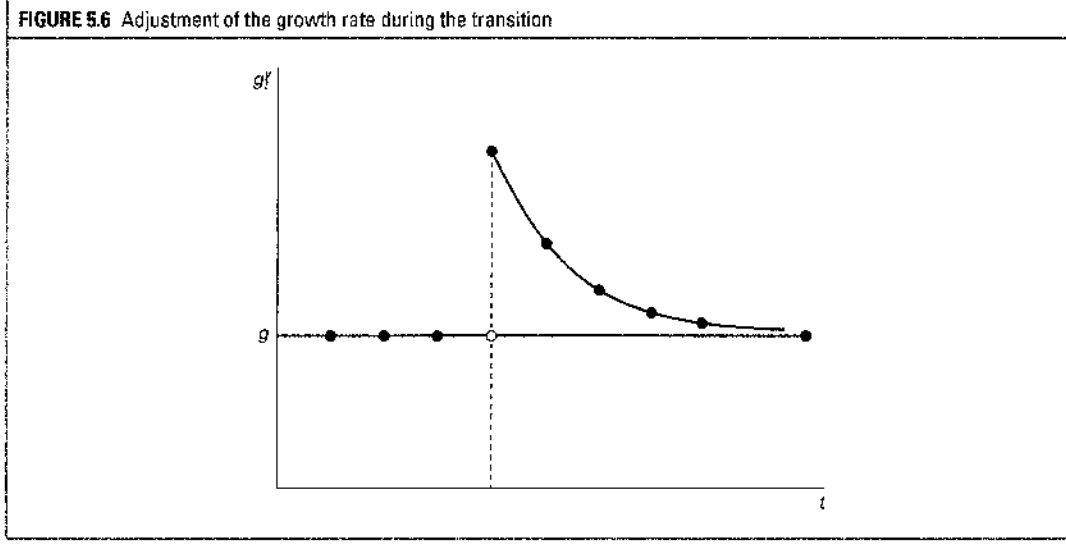
From the first period after the savings rate has increased, $\bar{k}_t$ begins to increase as illustrated in the Solow diagram as well as in the modified Solow diagram. From the modified Solow diagram, Fig. 5.4, one can see that the growth rate, $(\bar{k}_{t+1} - \bar{k}_t)/\bar{k}_t$ of $\bar{k}_t$ initially jumps up to a positive level and then decreases monotonically back towards zero. When $\bar{k}_t = k_t/A_t$ increases, the numerator, k_t , must be increasing by a rate larger than the rate by which the denominator, A_t , increases, that is, by a rate larger than g . The approximate growth rate of k_t is $g_t^k \equiv \ln k_t - \ln k_{t-1} = (\ln \bar{k}_t - \ln \bar{k}_{t-1}) + (\ln A_t - \ln A_{t-1}) \equiv g_t^{\bar{k}} + g_t^A$, so any relative increase in $\bar{k}_t$ is equal to a relative increase in k_t in excess of the relative increase in A_t . As long as k_t increases faster than A_t , so does y_t , since $g_t^y = \alpha g_t^k + (1 - \alpha)g_t^A$.

Along the adjustment to the new steady state, $\bar{k}^{**}$ and $\bar{y}^{**}$, represented by the upper straight line in Fig. 5.5, the capital intensity must be increasing all the time at a rate which is larger than g , but gradually falls back towards g . The same must then be true for y_t . The transition from the old to the new steady state is illustrated by the curve going from the lower to the upper straight line in Fig. 5.5. Figure 5.6 shows the adjustment of the growth rate of y_t during the transition from the old to the new steady state.

In the basic Solow model an increase in the savings rate implied that in the long run output per worker, y_t , changed from one constant level to a new and higher constant level. In the Solow model with technological progress the growth path of y_t shifts from one level to a new and higher level, with the long-run growth rate being the same before and after, namely g , as illustrated in Fig. 5.5. This happens through a transition where the growth rate of y_t first jumps up above g , and then falls monotonically back towards g , as shown in Fig. 5.6. One particular case is when $g = 0$. Then the growth paths of Fig. 5.5 will be horizontal, and the growth jump in Fig. 5.6 will occur around a long-run growth rate of zero. We are then back in the basic Solow model.

5.3 Steady state in the general Solow model

As we have seen, the steady state of the Solow model accords with the concept of balanced growth in *some* respects: the growth rates of capital per worker and of output per worker are constant and equal to each other. Furthermore, if $g > 0$, both growth rates are

FIGURE 5.6 Adjustment of the growth rate during the transition

positive. We are going to show that the steady state is in accordance with balanced growth in *all* respects.

The key endogenous variables in steady state

The technology-adjusted capital intensity in steady state, $\tilde{k}^*$, is found by setting $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}$ in (16) or (19) and solving for $\tilde{k}$. This gives:

$$\tilde{k}^* = \left(\frac{s}{n + g + \delta + ng} \right)^{\frac{1}{1-\alpha}}. \quad (21)$$

From $\tilde{y}_t = \tilde{k}_t^\alpha$ we then obtain the corresponding steady state value of $\tilde{y}_t$:

$$\tilde{y}^* = \left(\frac{s}{n + g + \delta + ng} \right)^{\frac{\alpha}{1-\alpha}}. \quad (22)$$

It is not really the *technology-adjusted* variables, $\tilde{k}_t$ and $\tilde{y}_t$, we are interested in. From the definitions, $\tilde{k}_t \equiv k_t/A_t$ and $\tilde{y}_t \equiv y_t/A_t$, it follows that when the economy has reached its steady state, capital per worker and output per worker in period t will be $k_t^* = A_t \tilde{k}^*$ and $y_t^* = A_t \tilde{y}^*$, respectively, or:

$$k_t^* = A_t \left(\frac{s}{n + g + \delta + ng} \right)^{\frac{1}{1-\alpha}}, \quad (23)$$

$$y_t^* = A_t \left(\frac{s}{n + g + \delta + ng} \right)^{\frac{\alpha}{1-\alpha}}. \quad (24)$$

Since consumption per worker in any period is $c_t = (1-s)y_t$, the steady state consumption path is:

$$c_t^* = A_t (1-s) \left(\frac{s}{n + g + \delta + ng} \right)^{\frac{\alpha}{1-\alpha}}. \quad (25)$$

The steady state expressions for the three variables above are all of the form A_t times a constant, and hence they all grow at the same rate g as A_t does. Note that substituting $A_0(1+g)^t$ for A_t will trace the three steady state growth paths above back to parameters and initial values. The steady state evolutions of the rental rates follow from inserting $\bar{k}^*$ for $\bar{k}_t$ in (17) and (18), respectively:

$$r^* = \alpha \left(\frac{s}{n+g+\delta+ng} \right)^{-1}, \quad (26)$$

$$w_t^* = A_t(1-\alpha) \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}}. \quad (27)$$

In steady state the rate of return on capital is a constant, r^* , so the real interest rate, $r^* - \delta$, is constant as well. The real wage rate, w_t^* , increases with A_t at rate g .

In fact all the requirements for balanced growth, as defined in Section 2.4, are fulfilled in the steady state: GDP per worker, capital per worker, consumption per worker, and the real wage rate all grow by one and the same constant rate, g . The labour force grows at the constant rate, n , and therefore (from $Y_t = y_t L_t$, $C_t = c_t L_t$, $I_t = S_t = sy_t L_t$, $K_t = k_t L_t$), GDP, consumption, investment and capital all grow at one and the same constant rate, approximately $g + n$. Finally, as we have shown, the rate of return on capital and the real interest rate are constant.

STEADY STATE



The Solow model passes the fundamental empirical test that its steady state prediction accords with balanced growth, and at the same time allows a strictly positive growth rate of GDP per worker ($g > 0$). The elasticity of long-run income per capita with respect to the investment rate s is $\alpha/(1-\alpha)$, the exponent in (24), and the elasticity with respect to $n+g+\delta$ is $-\alpha/(1-\alpha)$. Both elasticities should be around one half in absolute value given a labour's share of around $\frac{2}{3}$.

Structural policy for steady state

The above expressions and conclusions for output per worker and consumption per worker in steady state are reminiscent of what was found in the basic Solow model of Chapter 3. The similarity between the models implies that some of their policy implications and the relevant empirical tests of their predictive power are also quite similar.

With respect to structural policy, the general Solow model points to one type that could not be considered in the basic Solow model: a policy to increase the growth rate of technology. It is not easy to see from the present model what kind of policy could achieve this, and we will (again) defer further discussion of policies to affect technology to later chapters that consider models with endogenous technical innovation.

Otherwise, policies suggested by the general Solow model to increase income per person, now intended to elevate the whole growth path of income per person, are mainly policies that can increase the savings rate or decrease the population growth rate and possibly the depreciation rate. Note also from (25) that the value of the savings rate that maximizes consumption per person (by taking the consumption growth path to the highest possible level) is still $s^{**} = \alpha$. The golden rule savings rate is unchanged from the basic Solow model.

Empirics for steady state

Taking A_t and g as given, the implication of (24) is that higher s and lower n will generate higher steady state output per worker. In Chapter 3 we tested this prediction in Fig. 3.7, plotting GDP per worker against gross investment rates across countries, and in Fig. 3.8, plotting GDP per worker against population growth rates. Thus we tested the influences of

s and n on GDP per worker separately and found the model's prediction of the *directions* of these influences confirmed by the data. However, Eq. (24) implies a more precise theory of the influences of s and n on y_t than just their directions (the same was true for the counterpart equation for the basic Solow model). Taking logs on both sides of (24) gives:

$$\ln y_t^* = \ln A_t + \frac{\alpha}{1-\alpha} [\ln s - \ln(n + g + \delta + ng)]. \quad (28)$$

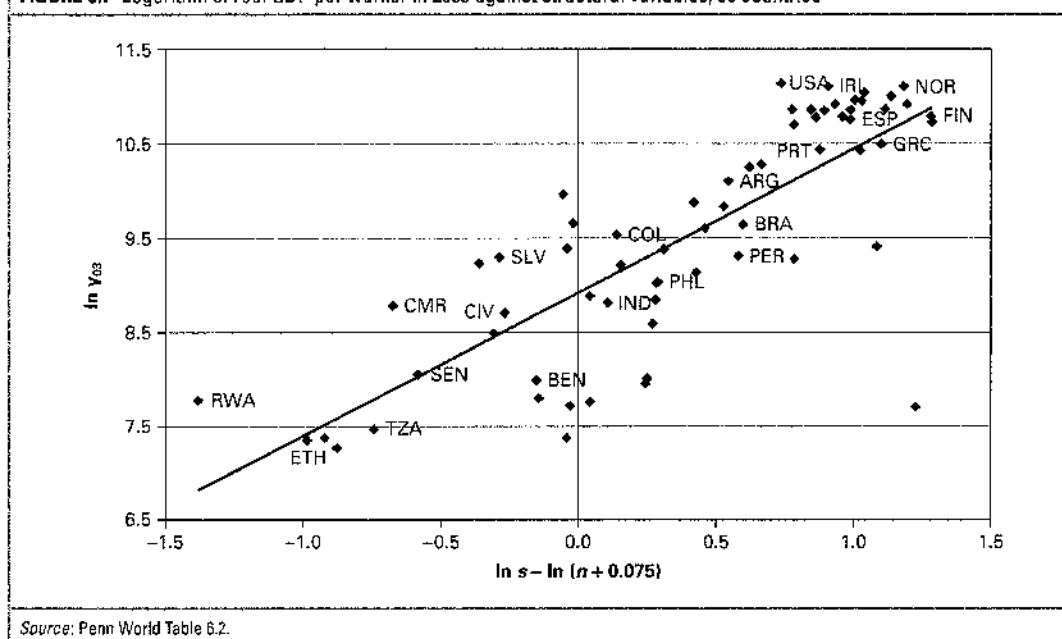
As far as empirics are concerned, this is the first of two important relationships resulting from the Solow model. This one concerns steady state, the other one (to be derived below) will concern convergence to steady state. Given the technological level, A_t , the steady state prediction of the model is that $\ln y_t^*$ should depend on $[\ln s - \ln(n + g + \delta + ng)]$, and the relationship should be a linear one with a positive slope equal to $\alpha/(1-\alpha)$. The slope should therefore be around $\frac{1}{2}$, since the capital share, α , is around $\frac{1}{3}$.

In Fig. 5.7 we have put this prediction to a test by considering the representative sample of 65 countries studied and described in association with Figs 2.5 and 2.6. We assume, perhaps heroically, that the countries considered were all in steady state in 2003 and had the same technological level, A_{03} , in that year. The figure plots, across countries i , the log of GDP per worker in 2003, $\ln y_{03}^i$, against $\ln s^i - \ln(n^i + 0.075)$, where s^i is the average gross investment rate in country i over the period 1960 to 2003, and n^i is the average labour force growth rate over the same period. We have thus set $g + \delta + ng$ at 7.5 per cent for all countries. Since ng is the product of two small growth rates, it is very small. Thus, without sacrificing much precision, we could have excluded ng in the above formulae. Hence we are assuming that $g + \delta \approx 0.075$, an estimate that is often used. (Constructing Fig. 5.7 we are more or less doing what you were asked to do yourself in Exercise 9 of Chapter 3.)

The figure is in fairly nice accordance with a positive and linear relationship. The straight line that has been drawn is a line of best fit resulting from an OLS estimation of the regression equation:

$$\ln y_{03}^i = \gamma_0 + \gamma [\ln s^i - \ln(n^i + 0.075)] + \xi^i, \quad (29)$$

FIGURE 5.7 Logarithm of real GDP per worker in 2003 against structural variables, 65 countries



where ξ^i is a random error term representing the influence of omitted variables (if necessary, consult the appendix on regression analysis regarding distributional assumptions on ξ^i etc.). The details of the estimation are:

$$\ln y_{03}^i = 8.92 + \frac{1.52}{(se = 0.14, t = 10.5)} [\ln s^i - \ln(n^i + 0.75)], \quad R^2 = 0.64,$$

The estimate of γ , the slope of the line, is 1.52. This slope is considerably larger than the 0.5 it should be according to our theory. Note that the standard error of the estimate of γ is 0.14. The 95 per cent confidence interval for the estimate goes approximately from two standard deviations below to two standard deviations above the estimate of γ , giving a confidence interval of [1.24; 1.80]. This is not close to including the value 0.5, so the estimated slope is *significantly* larger than 0.5. In fact, for the fraction $\alpha/(1 - \alpha)$ to equal 1.52, α would have to be 0.6. An α of around 0.6 is way above the capital share (around $\frac{1}{3}$) that α should correspond to.³

STEADY STATE AND THE CROSS-COUNTRY EMPIRICAL EVIDENCE



The steady state of the Solow model matches cross-country data quite well (even under an assumption of a common technological level): there is a clear positive correlation between $\ln y^i$ and $\ln s^i - \ln(n^i + 0.75)$ as predicted, and the linear form of the relationship is in nice accordance with the data. However, the model considerably underestimates the strength of the influences of the key parameters on GDP per person compared to the correlation found in the data.

The nice positive correlation in Fig. 5.7 is reassuring for our model, but it does not *prove* that the model's prediction of a positive long-run influence of the investment rate on income per capita is correct. In empirical analysis one must be careful not to confuse correlation with causation and about what is exogenous and what is endogenous. For instance, the picture would be consistent with a *reversed causality* where the exogenous variable was income per capita, the endogenous variable was the investment rate net of labour force growth, and income per capita had a positive influence on the investment rate net of labour force growth, perhaps through reducing population growth. It could also be *spurious correlation*, where there would be no causal relationship between the two in any direction, both would be endogenous, and a third exogenous factor, the 'entrepreneurial spirit' of a society, say, influenced both. Assume that entrepreneurial spirit is exogenous and affects both the investment rate and income per capita positively. Then more 'entrepreneurial' societies would tend to have higher investment rates *and* higher incomes per capita, so a positive correlation between the two would be observed without any causation between them. It may be, in fact, that there is indeed a positive influence of the investment rate net of labour force growth on income per worker *and* a positive influence of income per capita on the investment rate net of labour force growth *and* third factors influencing both systematically. In that case a picture like Fig. 5.7 will involve many different forces and viewed as a picture expressing the influence of the investment rate net of labour force growth on income per capita it will be highly distorted.

For now, just pick up that there is always a danger of misinterpreting correlation as causality. The positive correlation between the number of policemen and crime rates observed in many countries does not necessarily mean that more policemen imply more crime. It could be a simple reflection that more crime calls for more police.

3. The 95 per cent confidence interval for the implied α is [0.55; 0.64]. Show this.

5.4 Economic growth and convergence in the general Solow model⁴

It is reassuring that the Solow model can create balanced growth with a positive growth rate in steady state, but the model has transitory growth as well. As we have seen above, while the economy converges towards a steady state from below, the growth in GDP per worker will exceed the steady state growth rate, g , in such a way that the growth in excess of g decreases and goes to zero in the long run. The monotonically decreasing transitory growth means that the general Solow model accords with the idea of *conditional convergence* explained in Chapter 2. We will now examine the convergence process in detail.

Convergence according to the Solow model

Consider two countries that have the same structural characteristics, $\alpha, s, n, \delta, A_0, g$. They will then have identical modified Solow diagrams (see Fig. 5.4 for such a diagram). It follows that the country which is initially furthest below steady state will have the highest growth rate in $\tilde{k}_t$. As we showed above, this will imply that it has the highest growth rate in y_t as well. Through the process where each country approaches its own steady state, and does so faster the further away it is from steady state, the two countries will converge towards having the same level of GDP per worker.

Hence the Solow model supports conditional convergence. An implication is that if one controls appropriately for the structural differences between countries, there should be a negative relationship between GDP per worker in an initial year and the growth rate in subsequent years. In Chapter 2 we began to test the idea of convergence empirically by setting up growth in GDP per worker against the initial level of GDP per worker across countries, but there were things we could not do. When we wanted to control for structural differences between countries, we had to *postulate* an equation stating how the structural characteristics should enter (Eq. (4) of Chapter 2). Now that we have a real growth model we can explain how Eq. (4) is rooted in theory.

Furthermore, just as important as convergence itself is the *rate of convergence*, that is, the speed by which a country converges towards its long-run growth path. For a poor country that attempts to improve its structural characteristics (e.g., the savings rate) to approach a new and higher growth path, it is a matter of great concern whether it takes 10 or 50 years to go halfway to the new growth path, since this may be decisive for how fast poverty can be fought. In Chapter 2 we could not estimate rates of convergence, but now that we have a real growth model we can.

From the Solow model we will derive a ‘convergence equation’. This will state how a country’s average growth rate in GDP per worker over a period should depend on its structural characteristics and on its initial level of GDP per worker, that is, the Solow model’s convergence equation will rationalize (4) of Chapter 2. Furthermore, the convergence equation will have a coefficient on the initial GDP per worker that is linked one-to-one to the rate of convergence, so an estimate of this coefficient will provide us with an estimate of the rate of convergence. Here is how to proceed.

Start from the transition equation (16). Write it in the generic form, $\tilde{k}_{t+1} = G(\tilde{k}_t)$, where the function $G(\tilde{k}_t)$ is given as the expression on the right-hand side of (16). The steady state value, $\tilde{k}^*$, stated in (23) fulfils $\tilde{k}^* = G(\tilde{k}^*)$. We now *linearize the transition equation around the steady state*. For any differentiable function $f(x)$, and for any given $\bar{x}$, we have $f(z) - f(\bar{x}) \approx f'(\bar{x})(z - \bar{x})$, as long as z is not too far from $\bar{x}$. This is really just using the definition of the

4. Some readers may find this section more difficult than the material presented up to now. It is possible to skip this section, and a corresponding one in Chapter 6, and still read the remainder of the book. It would be a pity, though, since these sections are concerned with convergence to steady state, which is just as important an element of our theory as steady state itself. The only additional mathematical element used in this section is how to solve a first-order linear difference equation. For those who do not yet know the technique, we will give a cookbook recipe for the solution.

derivative, $f'(\bar{x})$. If $\bar{x}$ is such that $\bar{x} = f(\bar{x})$, then we get $f(z) - \bar{x} \approx f'(\bar{x})(z - \bar{x})$. Using these properties for the dynamic equation, $G(\bar{k}_t)$, measuring at the steady state value, $\bar{k}^*$, we obtain:

$$\bar{k}_{t+1} - \bar{k}^* = G'(\bar{k}^*)(\bar{k}_t - \bar{k}^*). \quad (30)$$

This is a new dynamic equation, a first-order difference equation in $\bar{k}_t$ (or in the *absolute* discrepancy between $\bar{k}_t$ and its steady state value). As long as we are not too far from steady state, the new dynamic equation approximately represents (16). Importantly, the new difference equation is linear, which means that we can solve it. Before doing so we will take it through some rewriting, but the linear structure, and hence the solvability, will be preserved.

The linearized dynamic equation includes $G'(\bar{k}^*)$. We can derive an expression for $G'(\bar{k}^*)$ in terms of model parameters by differentiating the right-hand side of (16) with respect to $\bar{k}_t$, and then evaluating at $\bar{k}_t = \bar{k}^*$, where $\bar{k}^*$ is given in (21):

$$\begin{aligned} G'(\bar{k}_t) &= \frac{1}{(1+n)(1+g)} [s\alpha(\bar{k}_t)^{\alpha-1} + (1+\delta)] \Rightarrow \\ G'(\bar{k}^*) &= \frac{1}{(1+n)(1+g)} [\alpha(n+g+\delta+ng) + (1+\delta)]. \end{aligned} \quad (31)$$

In the following we will still use the term $G'(\bar{k}^*)$ for short, but $G'(\bar{k}^*)$ has now been expressed in terms of model parameters, which can be used whenever wanted. From (31) it is easy to demonstrate that $0 < G'(\bar{k}^*) < 1$ (do that). This implies that the linearized dynamic equation (30) is stable, so $\bar{k}_t \rightarrow \bar{k}^*$ as $t \rightarrow \infty$ according to it.

Our next step is to convert (30) into changes in natural log-values rather than changes in absolute values. This is easily done using (again) $f(z) - f(\bar{x}) \approx f'(\bar{x})(z - \bar{x})$, this time letting the function $\ln \bar{k}_t$ take the place of f , and measuring at $\bar{k}^*$: $\ln \bar{k}_t - \ln \bar{k}^* = (1/\bar{k}^*)(\bar{k}_t - \bar{k}^*)$, or $\bar{k}_t - \bar{k}^* = \bar{k}^*(\ln \bar{k}_t - \ln \bar{k}^*)$. Using this on both sides of (30) gives (note that $\bar{k}^*$ cancels on both sides):

$$\ln \bar{k}_{t+1} - \ln \bar{k}^* = G'(\bar{k}^*)(\ln \bar{k}_t - \ln \bar{k}^*). \quad (32)$$

We convert into log-differences of $\bar{y}_t$ rather than $\bar{k}_t$, applying that from $\bar{y}_t = \bar{k}_t^\alpha$, one has $\ln \bar{y}_t = \alpha \ln \bar{k}_t$, and then $\ln \bar{y}_t - \ln \bar{y}^* = \alpha(\ln \bar{k}_t - \ln \bar{k}^*)$. Using this on both sides of (32) gives:

$$\ln \bar{y}_{t+1} - \ln \bar{y}^* = G'(\bar{k}^*)(\ln \bar{y}_t - \ln \bar{y}^*). \quad (33)$$

This is still a linear, first-order difference equation, now in $\ln \bar{y}_t$, and since $0 < G'(\bar{k}^*) < 1$, it implies that $\bar{y}_t \rightarrow \bar{y}^*$ as $t \rightarrow \infty$. The advantage of having transformed it into log-differences (or *relative* discrepancies between $\bar{y}_t$ and its steady state value) is that the rate of convergence appears in a well-defined way. To see this, first subtract $\ln \bar{y}^*$ and add $\ln \bar{y}^*$ on both sides of (33) to obtain:

$$\ln \bar{y}_{t+1} - \ln \bar{y}_t = (1 - G'(\bar{k}^*))(\ln \bar{y}^* - \ln \bar{y}_t). \quad (34)$$

Here we define $\lambda \equiv 1 - G'(\bar{k}^*)$, where we must have $0 < \lambda < 1$, since $0 < G'(\bar{k}^*) < 1$. Equation (34) then reads $\ln \bar{y}_{t+1} - \ln \bar{y}_t = \lambda(\ln \bar{y}^* - \ln \bar{y}_t)$, saying that in any period t , the relative change in $\bar{y}_t$ (from period t to $t+1$, the left-hand side) is the fraction λ of the current relative deviation between $\bar{y}_t$ and its long-run equilibrium level, $\bar{y}^*$. In other words, the remaining relative gap, $\ln \bar{y}^* - \ln \bar{y}_t$, is reduced by the fraction λ in any period, irrespective of how large the remaining gap currently is. Hence, λ is a time-independent measure of the rate at which $\bar{y}_t$ converges to $\bar{y}^*$, and we call λ the 'rate of convergence'. For instance, if λ is 0.02, then each year 2 per cent of the remaining distance to steady state is covered. Note that (34) has the 'convergence property': the growth rate of $\bar{y}_t$ is larger (in absolute value) the further away from steady state $\bar{y}_t$ is.

From (31) we can express the rate of convergence in terms of parameters as:

$$\lambda \equiv 1 - G'(\bar{k}^*) = \frac{1}{(1+n)(1+g)}(1-\alpha)(n+g+\delta+ng) \approx (1-\alpha)(n+g+\delta). \quad (35)$$

For the latter approximation we have used the facts that since both n and g are small, the fraction $1/[(1+n)(1+g)]$ is close to one, and ng is close to zero. Thus we may conclude that, according to the Solow model, the rate of convergence is approximately $\lambda = (1-\alpha)(n+g+\delta)$. For realistic parameter values such as $\alpha \approx \frac{1}{3}$ and $n+g+\delta \approx 0.075$ our model predicts a convergence rate of 5 per cent. A reasonable lower-bound estimate for λ would be around 4 per cent, e.g., corresponding to $\alpha = \frac{1}{3}$ and $n+g+\delta = 0.06$.

Inserting λ for $1 - G'(\bar{k}^*)$ in (34) and rearranging a bit gives:

$$\ln \tilde{y}_{t+1} = \lambda \ln \tilde{y}^* + (1-\lambda) \ln \tilde{y}_t, \quad (36)$$

which is a standard way of writing our linear, first-order difference equation in $\ln \tilde{y}_t$. To arrive at the Solow model's convergence equation we have to solve (36), that is, determine how $\ln \tilde{y}_t$ must depend on time t for (36) to be fulfilled in all periods. This is done using a standard 'cookbook procedure' involving the following steps.

First, one particular solution (a time path for $\ln \tilde{y}_t$ that makes (36) hold for all years) is the constant one: $\ln \tilde{y}_t = \ln \tilde{y}^*$ (check this).

Second, the so-called homogeneous equation, arising from setting the constant term ($\lambda \ln \tilde{y}^*$) in (36) equal to zero, is: $\ln \tilde{y}_{t+1} - (1-\lambda) \ln \tilde{y}_t = 0$. The characteristic polynomial associated with this is: $Q(x) = x - (1-\lambda)$, with root $x = 1-\lambda$ (the x that implies $Q(x) = 0$). The complete solution for the homogeneous difference equation is then: $\ln \tilde{y}_t = C(1-\lambda)^t$, where C is an arbitrary constant.⁵

Third, the complete solution to (36) is the sum of a specific solution and the complete solution for the homogeneous equation: $\ln \tilde{y}_t = \ln \tilde{y}^* + C(1-\lambda)^t$.⁶

Fourth, for the solution to be compatible with a given initial value $\tilde{y}_0$ for $t = 0$, we must have $C = \ln \tilde{y}_0 - \ln \tilde{y}^*$. Therefore, the unique solution to (36), given the initial value $\tilde{y}_0$, is:

$$\begin{aligned} \ln \tilde{y}_t &= \ln \tilde{y}^* + (\ln \tilde{y}_0 - \ln \tilde{y}^*)(1-\lambda)^t \\ &= [1 - (1-\lambda)^t] \ln \tilde{y}^* + (1-\lambda)^t \ln \tilde{y}_0. \end{aligned} \quad (37)$$

Evaluating for a specific year T and subtracting $\ln \tilde{y}_0$ on both sides gives:

$$\ln \tilde{y}_T - \ln \tilde{y}_0 = [1 - (1-\lambda)^T](\ln \tilde{y}^* - \ln \tilde{y}_0).$$

Now using $\ln \tilde{y}_t = \ln y_t - \ln A_t$ for both $t = T$ and $t = 0$, and dividing by T on both sides gives:

$$\frac{\ln y_T - \ln y_0}{T} = \frac{\ln A_T - \ln A_0}{T} + \frac{1 - (1-\lambda)^T}{T} (\ln A_0 + \ln \tilde{y}^* - \ln y_0). \quad (38)$$

Here we can (finally) insert from (24) how the steady state value $\tilde{y}^*$ depends on the structural characteristics. When we do that and also insert that $(\ln A_T - \ln A_0)/T$ is approximately equal to g , we get:

$$\frac{\ln y_T - \ln y_0}{T} \approx g + \frac{1 - (1-\lambda)^T}{T} \left(\ln A_0 + \frac{\alpha}{1-\alpha} [\ln s - \ln(n+g+\delta+ng)] - \ln y_0 \right). \quad (39)$$

This is the convergence equation of the Solow model. You may think that it looks complicated, but its content is simple: on the left-hand side is the average annual growth rate in GDP per

5. Check that $\ln \tilde{y}_t = C(1-\lambda)^t$ fulfils the homogeneous equation for all t , irrespective of C .

6. Check again, this time that the stated solution fulfils (36) for all t irrespective of C .

worker over a period from year zero to year T . This growth rate is, according to (39) (or (38)), given as the sum of two terms: the first is simply the growth rate, g , along the steady state growth path, $A_0 \tilde{y}^*$, for y_t . The other depends increasingly on the initial relative gap (the year zero log-difference) between the steady state growth path, taking the value $A_0 \tilde{y}^*$ in year zero, and the actual GDP per worker, y_0 . The dependence is such that as the number of years considered becomes large, as T goes to infinity, the second term will contribute very little to average growth. This is intuitively as it should be for an economy monotonically approaching an own long-run growth path with a constant growth rate, g .

CONVERGENCE THEORETICALLY



Convergence property: Income per worker converges towards a steady state growth path along which the growth rate is g , and which is fully characterized by the structural variables, s , n , g , δ , and A_0 . The growth rate of income per worker is larger the lower the current level of income per worker.

Speed of convergence: The rate of convergence (to steady state), λ , is approximately $(1 - \alpha)(n + g + \delta)$, which on an annual basis should be in the range of 4 to 6 per cent for plausible parameter values. Each year 4 to 6 per cent of the remaining gap to steady state should thus be eliminated.

Testing the model outside steady state and estimating the rate of convergence

The theoretical convergence equation, (39), and an assumption that countries are technologically similar, having the same g and A_0 , suggest a regression equation across countries i :

$$g_{T,0}^i \equiv \frac{\ln y_T^i - \ln y_0^i}{T} = \beta_0 - \beta_1 \ln y_0^i + \beta_2 [\ln s^i - \ln(n^i + 0.075)], \quad (40)$$

where this time we do not bother to write an error term like the ξ^i of (29) explicitly. Eq. (40) is exactly the same as (4) of Chapter 2.⁷ From appropriate data we can estimate the parameters β_0 , β_1 , and β_2 by OLS. We do that for the same sample of 65 countries considered in Figs 2.5 and 2.6, where, for example, s^i is the average investment rate between 1960 and 2003 of country i , etc. In fact, to create Fig. 2.6, we already did this estimation. The details of the estimation, with standard errors in parentheses below estimates, are:

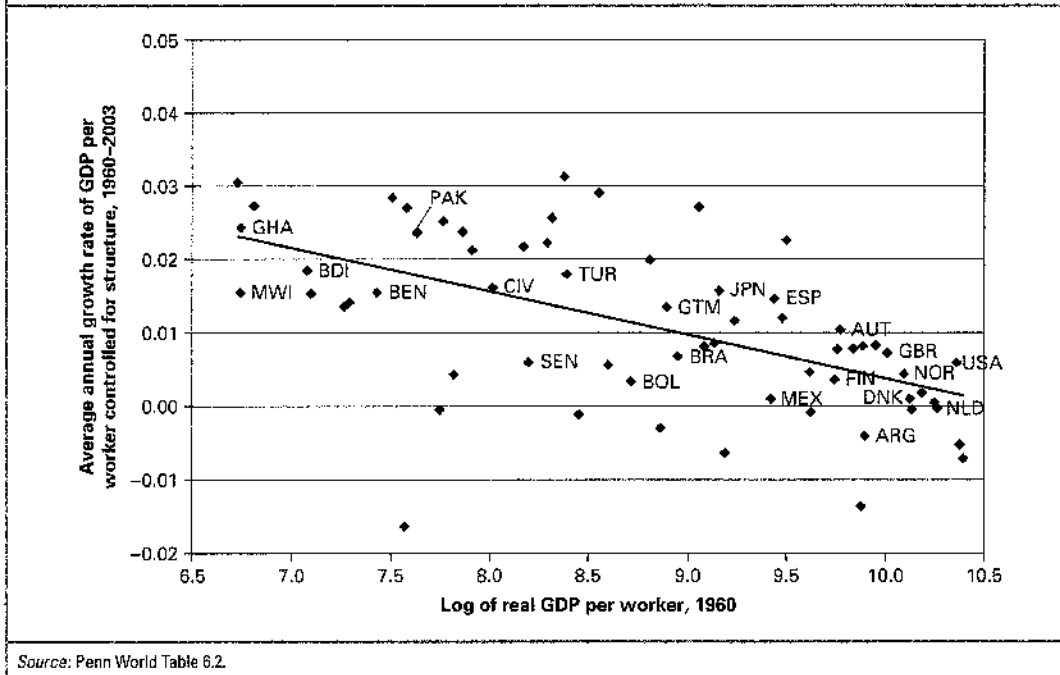
$$g_{03,60}^i = 0.063 - \underset{(se=0.0015)}{0.006 \ln y_{60}^i} + \underset{(se=0.0023)}{0.016 [\ln s^i - \ln(n^i + 0.075)]}, \quad \text{adj. } R^2 = 0.38. \quad (41)$$

This is why we chose the parameter value $\beta_2 = 0.016$ in the construction of Fig. 2.6. We were then plotting $(g_{03,60}^i - 0.016[\ln s^i - \ln(n^i + 0.075)])$ against y_{60}^i , and we indicated the straight line, $0.063 - 0.006 \ln y_{60}^i$, resulting from our estimation. For convenience the figure is repeated here as Fig. 5.8.

The figure and the details of the estimation reported in (41) show that the explanatory power of our regression equation is not bad. Even under an assumption that the many countries considered are technologically similar, the model's convergence equation stands up quite well when confronted with cross-country data. One may take this as evidence that

7. In specifying this regression equation we are making a short-cut: according to (39) the parameters β_0 , β_1 , and β_2 in (36) all depend on λ , which in turn depends on n as stated in (35). Nevertheless, we hold β_0 , β_1 , and β_2 constant across countries in our estimations, ignoring that cross-country differences in n^i should imply cross-country differences in λ^i . We simply postulate a common rate of convergence λ and see how good our estimations become. The reason is that if we accounted for the influence of n on λ , the resulting non-linear equation would require more sophisticated methods to estimate.

FIGURE 5.8 Average annual growth rate of GDP per worker adjusted for structural characteristics against the initial level of GDP per worker, 65 countries



the assumption that countries have equal access to technology is not as far-fetched as it may first seem, an issue we will return to in the next chapter where the convergence equation derived will perform even better.

Importantly, our estimation in (41) implies an estimate of the rate of convergence. According to (39), the parameter β_1 is equal to $(1 - (1 - \lambda)^T)/T$, implying that $\lambda = 1 - (1 - T\beta_1)^{1/T}$. Inserting $T = 43$ and our estimate of β_1 of 0.006 gives $\lambda = 0.007$. Thus our estimate of the annual rate of convergence based on our model and data for the 'countries of the world' is 0.7 per cent. Taking uncertainty into consideration, the 95 per cent confidence interval for λ goes from approximately 0.3 per cent to 1.1 per cent, computed (again) the simple way: go two standard errors down and two standard deviations up from the estimated β_1 to find (approximately) the 95 per cent confidence interval [0.003; 0.009] for β_1 . Then insert the two end points of the interval into the formula for λ to find the 95 per cent confidence interval for λ . (Do this yourself.)

The estimated annual rate of convergence is substantially smaller than what it should be according to the model for realistic parameter values. Above we found that for plausible parameter values, predicted annual rates of convergence should be no less than 4 per cent. This suggests that the Solow model exaggerates considerably the speed of convergence compared to real-world convergence.

CONVERGENCE EMPIRICALLY



When confronted with real-world cross-country data, the general Solow model stands up quite well with respect to its out-of-steady-state convergence prediction: one does find a relatively clear negative relationship between the initial value of and the subsequent growth in income per worker across the countries of the world. However, the estimated speed of convergence is significantly lower than the one derived from the theoretical model.

In summary, the general Solow model performs well confronted with real-world data, both with respect to its steady state *level* prediction and with respect to its out-of-steady-state *convergence* prediction. For both of these predictions there is, however, a problem of magnitudes. In Chapter 6 we will present an extension of the Solow model that implies substantial improvements with respect to both of these empirical shortcomings.

5.5 Growth accounting

Growth accounting was suggested by Robert Solow in a paper that followed up on his famous 1956 article.⁸ Growth accounting uses just one of the equations of the Solow model, the aggregate production function. We will still assume the production function to be of the Cobb–Douglas form. In growth accounting exercises one often expresses technical progress as (Hicks-neutral) changes in the total factor productivity, B_t , but one could equally well express it as labour-augmenting changes in an appropriately defined A_t , remembering that $A_t = B_t^{1/(1-\alpha)}$. So the aggregate production of a country in year t is assumed to depend on the inputs of capital and labour and on the level of technology as:

$$Y_t = B_t K_t^\alpha L_t^{1-\alpha} = K_t^\alpha (A_t L_t)^{1-\alpha}. \quad (42)$$

For another and later year $T > t$, we have:

$$Y_T = B_T K_T^\alpha L_T^{1-\alpha}. \quad (43)$$

If we take logs on both sides in each of the two above equations, subtract the first from the second, and divide on both sides by $T - t$, we get:

$$\frac{\ln Y_T - \ln Y_t}{T - t} = \frac{\ln B_T - \ln B_t}{T - t} + \alpha \frac{\ln K_T - \ln K_t}{T - t} + (1 - \alpha) \frac{\ln L_T - \ln L_t}{T - t}. \quad (44)$$

The left-hand side is the approximate average annual growth rate of GDP between t and T . The right-hand side is the approximate average annual growth rate of total factor productivity plus a weighted average of the approximate growth rates of capital and labour. Equation (44) thus splits up the growth in GDP into growth components.

If one has data for GDP and for the inputs of capital and labour for two years t and T , and if one sets the capital share α at $\frac{1}{3}$, (44) can be used to compute the contributions to total growth from each of the growth components. In particular, the contribution, $(\ln B_T - \ln B_t)/(T - t)$, of growth in the total factor productivity can be computed residually. This component is called the *Solow residual*.

How informative is a calculation of the average annual growth rate in total factor productivity of, say 3 per cent, over some period? In a way, not very informative, since we do not really know what B_t is. We have simply put all factors of importance for aggregate production other than physical capital and labour into B_t . Solow himself has expressed this by saying that the Solow residual may be viewed as ‘a measure of our ignorance’. However, although we do not know exactly what an increase in B_t represents, growth accounting exercises may be useful. For instance, one could compare the Solow residuals between two different periods for one country, or between two different countries for the same period. If one residual is considerably larger than the other, the contribution to growth coming from unknown factors has changed from one period to the other, or has been larger in one country than in the other. Lower growth in B_t may be a first warning that productivity is not developing as favourably as it used to, or as favourably as in some other country.

8. The follow-up paper is: Robert M. Solow, ‘Technical Change and the Aggregate Production Function’, *Review of Economics and Statistics*, 39, 1957, pp. 312–320.

One can also do growth accounting per worker. The production function (42) implies that $y_t = B_t k_t^\alpha$ in the usual notation. Taking logs and time differences, etc., gives:

$$\frac{\ln y_T - \ln y_t}{T - t} = \frac{\ln B_T - \ln B_t}{T - t} + \alpha \frac{\ln k_T - \ln k_t}{T - t}. \quad (45)$$

For $B_t = A_t^{1-\alpha}$, this equation may be rewritten in terms of A_t to give

$$\frac{\ln y_T - \ln y_t}{T - t} = (1 - \alpha) \frac{\ln A_T - \ln A_t}{T - t} + \alpha \frac{\ln k_T - \ln k_t}{T - t}, \quad (46)$$

expressing the growth rate of y_t as the weighted average of the growth rates of A_t and k_t , in accordance with (5) above. Equation (45) splits up the growth in GDP per worker into contributions from growth in capital per worker and growth in total factor productivity. Again, from data on GDP per worker and capital per worker, and setting $\alpha = \frac{1}{3}$, one can determine the growth contribution from technology (or other factors) residually.

Table 5.1 reports on a growth accounting exercise for six different countries. For each country and for five-year periods between 1960–65 and 1995–2000, the average annual growth rate in GDP per worker has been split up in components according to (45) assuming $\alpha = \frac{1}{3}$, that is, according to $g^y = g^B + g^k/3$ in our usual notation.

Consider first the figures for the USA. As you will see, over the full period from 1960 to 2000, capital and GDP (per worker) grew at almost the same rate. This indicates that the USA was close to balanced growth over the period (and so was Belgium, while for the other countries the capital–output ratio either decreased or increased substantially). In the first five-year period TFP (total factor productivity) growth in the USA was 2.9 per cent per year, which is high. To illustrate, note that a g^B of 2.9 per cent (per year) corresponds to g^A of 4.4 per cent for $\alpha = \frac{1}{3}$ (from $A_t = B_t^{1/(1-\alpha)}$ it follows that $g^A = g^B/(1 - \alpha)$, which you should show), and under balanced growth, g^A is directly comparable to g^y . Annual growth rates in GDP per worker of around 4.4 per cent are very high and unusual as averages over longer periods in Western, early industrialized countries. The internationally comparable data used for Table 5.1 only go back to 1960, but for the USA one can go further back, and doing so will reveal that high TFP growth was typical for a longer period following the Second World War.

The period of high TFP growth in the USA was followed by a considerable slowdown with quite low rates of TFP growth in the late 1960s and the 1970s, followed again by a partial return to higher TFP growth in the 1980s and 1990s. The downward movement in TFP growth in the middle of the period has been called the ‘great productivity slowdown’, and you will see from Table 5.1 that the UK, Belgium, Sweden and Denmark experienced similar slowdowns, while Ireland seems to have been developing quickly throughout the period. Economists have searched for reasons for the international productivity slowdown described.

Some have suggested that the oil crises of 1973 and 1979 were responsible (but note that the slowdown in the USA started before that). Others have stressed the relative shift in the production mix away from manufacturing towards the more labour-intensive production of services where productivity gains may be harder to achieve. Still others have claimed that declines in the quality of the labour force or of infrastructure caused the slowdown. Moreover, some have argued that it is not the low TFP growth of the 1970s and 1980s that needs explanation. Rather it is the high TFP growth in the 1960s that is unusual and needs to be explained. There are pros and cons for all these hypotheses. What we want to emphasize is that the evidence from a growth accounting exercise has induced economists to think hard about what was going on. This illustrates that growth accounting is a useful tool for economic analysis.

Growth accounting only uses a production function and some knowledge of parameters (like our α). The production function does not have to be as simple as (42) above. It is often argued that growth accounting should allow not only for the *quantity* of labour input, measured in hours or person-years, but also for the *quality* of labour input, measured by

TABLE 5.1 Growth accounting: the great productivity slowdown

USA(%)					UK(%)				
Period	g^k	g^y	$g^k/3$	$g^B = g^y - g^k/3$	Period	g^k	g^y	$g^k/3$	$g^B = g^y - g^k/3$
1960-65	2.0	3.5	0.7	2.9	1960-65	—	3.0	—	—
1965-70	2.4	1.2	0.8	0.4	1965-70	-5.8	2.7	-1.9	4.6
1970-75	1.2	0.2	0.4	-0.3	1970-75	2.3	1.6	0.8	0.9
1975-80	1.0	1.0	0.3	0.7	1975-80	1.5	1.1	0.5	0.6
1980-85	1.7	1.6	0.6	1.0	1980-85	1.1	1.3	0.4	0.9
1985-90	0.8	1.5	0.3	1.2	1985-90	2.5	2.8	0.8	1.7
1990-95	1.0	1.3	0.3	1.0	1990-95	3.2	1.8	1.1	0.8
1995-00	2.2	2.5	0.7	1.7	1995-00	3.6	2.6	1.2	1.4
1960-00	1.5	1.6	0.5	1.1	1965-00	1.2	2.0	0.4	1.6

Ireland(%)					Belgium(%)				
Period	g^k	g^y	$g^k/3$	$g^B = g^y - g^k/3$	Period	g^k	g^y	$g^k/3$	$g^B = g^y - g^k/3$
1960-65	-0.8	3.6	-0.3	3.9	1960-65	1.2	4.2	0.4	3.8
1965-70	1.2	4.6	0.4	4.2	1965-70	2.5	4.3	0.8	3.5
1970-75	0.9	3.7	0.3	3.4	1970-75	2.7	2.7	0.9	1.8
1975-80	2.4	3.4	0.8	2.6	1975-80	2.3	2.4	0.8	1.7
1980-85	1.9	1.2	0.6	0.6	1980-85	2.7	1.1	0.9	0.2
1985-90	2.3	4.6	0.8	3.8	1985-90	2.7	2.6	0.9	1.7
1990-95	0.3	2.8	0.1	2.7	1990-95	2.2	0.9	0.7	0.2
1995-00	1.1	5.5	0.4	5.2	1995-00	2.1	2.0	0.7	1.3
1960-00	1.2	3.7	0.4	3.3	1960-00	2.3	2.5	0.8	1.8

Sweden(%)					Denmark(%)				
Period	g^k	g^y	$g^k/3$	$g^B = g^y - g^k/3$	Period	g^k	g^y	$g^k/3$	$g^B = g^y - g^k/3$
1960-65	—	4.6	—	—	1960-65	—	3.9	—	—
1965-70	2.7	3.2	0.9	2.3	1965-70	6.2	2.2	2.1	0.1
1970-75	2.9	1.5	1.0	0.5	1970-75	3.1	0.8	1.0	-0.2
1975-80	1.8	0.3	0.6	-0.3	1975-80	2.3	1.2	0.8	0.4
1980-85	1.9	1.6	0.6	1.0	1980-85	1.3	0.6	0.4	0.2
1985-90	2.4	1.6	0.8	0.8	1985-90	2.6	0.4	0.9	-0.5
1990-95	2.7	1.7	0.9	0.8	1990-95	3.0	2.4	1.0	1.4
1995-00	2.6	3.0	0.9	2.1	1995-00	3.0	2.3	1.0	1.3
1965-00	2.4	1.9	0.8	1.0	1965-00	3.1	1.4	1.0	0.4

Sources: OECD, Economic Outlook Database; For Denmark 1960-1970, Penn World Table 5.6

the average level of education. It is natural to measure the average level of education in year t by the average number of years, u_t , that people in the year t labour force have spent on education (in school, say). The quality of labour would then be given by an increasing function $h(u_t)$, often referred to as a human capital function, and the aggregate production function would be modified to:

$$Y_t = B_t K_t^\alpha (h(u_t) L_t)^{1-\alpha}. \quad (47)$$

We cannot do growth accounting based on (47) until we know more about the function $h(u_t)$. Here some results from labour economics can be of use. Many labour economists have

studied the relationship between education and wages. One of their findings (most often based on microeconomic data) is that a specific *absolute* change in the amount of education, one more year in school, say, seems to give rise to a specific *relative* change in the wage rate.⁹ By assuming the functional form:

$$h(u_t) = \exp(\psi u_t), \quad \psi > 0,$$

we obtain an increasing human capital function with exactly this property (just take logs on both sides and differentiate with respect to u_t):

$$\frac{dh(u_t)}{h(u_t)} \bigg/ du_t = \psi.$$

That is, one additional year of education yields a certain relative increase, ψ , in the level of human capital independently of the initial u_t . Let us now try to estimate ψ . If labour is paid its marginal product, then (47) implies that the wage rate, w_t , should be:

$$w_t = (1 - \alpha)B_t K_t^\alpha h(u_t)^{1-\alpha} L_t^{-\alpha} = (1 - \alpha)B_t \left(\frac{K_t}{L_t}\right)^\alpha \exp((1 - \alpha)\psi u_t).$$

Taking logs on both sides and differentiating with respect to u_t gives (check this for yourself):

$$\frac{dw_t}{du_t} \bigg/ w_t = (1 - \alpha)\psi.$$

Thus a one-year increase in schooling gives a relative increase in pay equal to $(1 - \alpha)\psi$. A rather robust finding in empirical studies (mainly for the USA, but for some other countries as well) is that the percentage increase in pay generated by a one-year increase in schooling is around 7 per cent. If $(1 - \alpha)\psi$ is roughly equal to 0.07, and α is $\frac{1}{3}$, then ψ must be approximately 0.1 (10 per cent). With the production function:

$$Y_t = B_t K_t^\alpha (\exp(\psi u_t) L_t)^{1-\alpha}, \quad (48)$$

and knowledge of the parameters α and ψ , we can do growth accounting again. In per worker terms the appropriate formula following from (48) is:

$$\frac{\ln y_T - \ln y_t}{T - t} = \frac{\ln B_T - \ln B_t}{T - t} + \alpha \frac{\ln k_T - \ln k_t}{T - t} + (1 - \alpha)\psi \frac{u_T - u_t}{T - t}. \quad (49)$$

Derive this formula yourself and note that it is not the average *relative* change in schooling from year t to T that enters, but the average *absolute* change. From data on GDP per worker, capital per worker, and average years of schooling of the labour force in two different years t and T , one can again split up growth into the contributions from growth in capital per worker, growth in education, and the residual growth in TFP.

In particular, if a country has experienced a large increase in education over the period considered, one may tend to exaggerate the contribution from TFP growth if one does not include education in the growth accounting. This may be particularly relevant for an assessment of TFP growth in the growth miracle countries of East Asia. The East Asian tiger economies have typically experienced huge increases in educational effort. An exercise will ask you to try to get an impression of the TFP growth in the East compared to the West, taking education into account.

If one ends up finding that a certain country has had considerably higher TFP growth than another, controlling appropriately for differences in education, one can go on looking

9. A pioneering contribution is Jacob Mincer, *Schooling, Experience, and Earnings*, Columbia University Press, 1974.

for other neglected factors of production to explain the difference. The tiger economies do indeed tend to exhibit very high TFP growth even after one has controlled for education. One explanation which has been suggested is that over the periods most often considered, the West has experienced a larger shift of the production mix from manufacturing to services than East Asia. It is an open question whether, taking *all* relevant factors into account, East Asia has had above-average TFP growth, or whether its high growth is simply due to above-average factor accumulation.

A growth accounting exercise can split up growth in GDP or in GDP per worker into components, thus attributing total growth to various sources. However, this does not identify the *causes* of the growth. Here is why. Assume that the growth process of some economy can be described fully by the steady state of the Solow model with appropriate parameter values. In that case there will be positive growth in GDP per worker if and only if $g > 0$. In a causal sense, all growth in GDP per worker will then be rooted in the exogenous growth of the technological variable. On the other hand, if one conducts a growth accounting exercise, one will find that the growth in GDP per capita comes partly from technological growth and partly from growth in capital per worker. Both observations will be true. Without technological growth there would be no growth in GDP per worker, so technological growth is the ultimate source of long-run growth. However, when there is (steady state) growth in GDP per worker (and hence in technology), there will also be growth in capital per worker, so part of the growth observed comes from capital accumulation. An exercise will ask you to figure out exactly what a growth accounting exercise for an economy in Solow steady state will say.

5.6 Summary



1. In the basic Solow model the only source of long-run growth in economic activity is population (or rather labour force) growth. The basic Solow model cannot generate the long-run growth in GDP per capita that we observe in the data.
2. The general Solow model developed in this chapter generalizes the basic Solow model by introducing steady exogenous technological progress. This enables the model to generate long-run growth in GDP per capita. Technical progress may be labour-augmenting, increasing the efficiency of labour; it may be capital-augmenting, raising the productivity of capital, or it may take the form of an increase in total factor productivity. As long as the aggregate production function has the Cobb–Douglas form, all of these types of technological progress are equivalent and have the same implications for the evolution of the economy.
3. In the long run the general Solow model converges on a steady state with balanced economic growth, where total output, consumption, investment and the capital stock all grow at a rate equal to the sum of the exogenous growth rates of population and labour-augmenting productivity, where output per capita (worker), capital per capita, consumption per capita and the real wage rate all grow at the same rate as labour-augmenting productivity, and where the real interest rate is constant. Hence the long-run economic predictions of the general Solow model accord with the basic stylized facts about long-run growth. Outside steady state the general Solow model accords with the observation of conditional convergence.
4. The general Solow model implies that structural economic policies which succeed in raising the economy's savings rate or in reducing its rate of population growth will gradually take the economy to a higher steady state growth path characterized by a

higher level of steady state income and consumption. The general Solow model implies the same golden savings rule as the basic Solow model: long-run consumption per capita will be maximized when the savings rate equals the capital income share of GDP.

5. Structural policies which increase the savings rate or reduce the population growth rate cannot permanently raise the economy's growth rate. This can only be achieved via a policy that permanently raises the growth rate of factor productivity, but the general Solow model is silent about the factors determining technological progress.
6. Empirical data for a large sample of countries around the world confirm the steady state prediction of the Solow model, that real GDP per worker will tend to be higher, the higher the investment rate and the lower the population growth rate. However, for reasonable values of the capital income share of GDP, the theoretical Solow model underestimates the observed quantitative effects of these structural characteristics.
7. Outside the steady state the general Solow model predicts conditional convergence: controlling for cross-country differences in structural characteristics, a country's growth rate will be higher, the lower its initial level of real GDP per worker. The data for countries around the world support this prediction, but for reasonable parameter values the theoretical model significantly overestimates the rate (speed) at which economies converge.
8. Using an aggregate production function, one can estimate the contributions to aggregate output growth stemming from increases in factor inputs and from increases in total factor productivity. Such a decomposition of the overall growth rate is called growth accounting. In basic growth accounting, growth in output per worker can originate from growth in the capital stock per worker or from growth in total factor productivity. Still, in a causal sense the ultimate source of growth is technological progress, since long-run growth in capital per worker can only occur if there is continued growth in total factor productivity.

5.7 Exercises

Exercise 1. The general Solow model in continuous time

In continuous time the general Solow model consists of the seven equations:

$$Y = K^\alpha (AL)^{1-\alpha},$$

$$r = \alpha \left(\frac{K}{AL} \right)^{\alpha-1},$$

$$w = (1-\alpha) \left(\frac{K}{AL} \right)^\alpha A,$$

$$S = sY,$$

$$\dot{K} = sY - \delta K,$$

$$\frac{\dot{L}}{L} = n,$$

$$\frac{\dot{A}}{A} = g,$$

where the restrictions on the parameters α , s , n , g and δ are the same as in the model in discrete time, except that we do not have to assume $n > -1$ and $g > -1$. It is assumed that $n + g + \delta > 0$. We use (for any point t in time) the usual definitions, $k \equiv K/L$ and $y \equiv Y/L$, as well as the definitions, $\tilde{k} \equiv K/(AL)$ and $\tilde{y} \equiv Y/(AL)$.

1. Applying methods similar to those used in Exercise 2 of Chapter 3, show that the law of motion for $\tilde{k}$ following from the general Solow model in continuous time can be expressed in the Solow equation:

$$\dot{\tilde{k}} = s\tilde{k}^\alpha - (n + g + \delta)\tilde{k}.$$

Compare to (19) for the model in discrete time.

2. Illustrate the above Solow equation in a Solow diagram with $\tilde{k}$ along the horizontal axis. Demonstrate (from the Solow diagram) that from any initial value, $\tilde{k}_0 > 0$, capital per effective worker, $\tilde{k}$, will converge towards a specific steady state level, $\tilde{k}^*$, in the long run (as $t \rightarrow \infty$).
3. Compute the steady state values, $\tilde{k}^*$ and $\tilde{y}^*$, for capital per effective worker and income per effective worker. Compare to the parallel expressions found in this chapter for the model in discrete time. Also compute the steady state value and path (respectively), r^* and w^* , for the real factor prices and compare again to the discrete time model.
4. Show that (it follows from the model that) the growth rate in $\tilde{k}$ at any time is:

$$\frac{\dot{\tilde{k}}}{\tilde{k}} = s\tilde{k}^{\alpha-1} - (n + g + \delta).$$

Illustrate this in a modified Solow diagram. Show that the growth rate, $\dot{y}/y$, of output per worker at any time t is α times the growth rate of $\tilde{k}$ plus g . Assume that initially $\tilde{k}_0 < \tilde{k}^*$. Describe and explain qualitatively, in words, the transition to steady state. For instance, how do r and w develop along the transition, and when is the growth rate in y at its highest?

5. To solve the general Solow model analytically one can apply a trick as in Exercise 3 of Chapter 3. Define $z \equiv \tilde{k}^{1-\alpha}$. Show that z is the capital-output ratio, k/y . Then show that:

$$\dot{z} = (1 - \alpha)s - \lambda z, \tag{50}$$

where $\lambda \equiv (1 - \alpha)(n + g + \delta)$. Note that (50) is linear. Show that the steady state value of z is $z^* = (1 - \alpha)s/\lambda = s/(n + g + \delta)$. Solve (50) by finding the characteristic roots, etc., and show that (given an initial capital-output ratio z_0) the solution is:

$$\begin{aligned} z &= \frac{(1 - \alpha)s}{\lambda}(1 - e^{-\lambda t}) + z_0 e^{-\lambda t} \\ &= \frac{s}{n + g + \delta}(1 - e^{-\lambda t}) + z_0 e^{-\lambda t}. \end{aligned}$$

This says that at any time t , the capital-output ratio is the weighted average of its steady state value and its initial value, with the former variable carrying an increasing weight over time.

Show that the solutions for $\tilde{k}$ and $\tilde{y}$ are:

$$\begin{aligned} \tilde{k} &= \left(\frac{s}{n + g + \delta}(1 - e^{-\lambda t}) + \tilde{k}_0^{1-\alpha} e^{-\lambda t} \right)^{1/(1-\alpha)}, \\ \tilde{y} &= \left(\frac{s}{n + g + \delta}(1 - e^{-\lambda t}) + \tilde{y}_0^{\frac{1-\alpha}{\alpha}} e^{-\lambda t} \right)^{\frac{\alpha}{1-\alpha}}. \end{aligned}$$

From these expressions, argue that λ is a measure of the speed of convergence and relate it to what was found for the rate of convergence (by considering a linear approximation around steady state) in Section 5.4.

Exercise 2. Analysing the Solow model in discrete time in terms of the capital–output ratio

(This exercise may be a useful preparation for Chapter 7, technically.) When analysing the general Solow model in this chapter, we were looking for some variables that would be constant in a steady state with balanced growth. We found that $\tilde{k}_t$ and $\tilde{y}_t$ had this property, and we analysed the model in terms of these variables. In steady state the capital–output ratio, $z_t \equiv K_t/Y_t$, must also be constant for the steady state to exhibit balanced growth.

1. Show that z_t as just defined fulfils: $z_t = k_t/y_t = \tilde{k}_t/\tilde{y}_t = \tilde{k}_t^{1-\alpha}$ with definitions as in this chapter, and show from the results of Section 3 that indeed z_t is constant in steady state and equal to:

$$z^* = \frac{s}{n + g + \delta + ng}.$$

This gives the idea that the model could alternatively have been analysed in terms of z_t . In fact, the previous exercise showed that analysing the Solow model in continuous time in terms of the capital–output ratio had the advantage that the law of motion was a *linear* differential equation that could be solved. This exercise asks you to analyse the model in discrete time in terms of z_t , among other reasons to see whether the law of motion becomes a linear, and therefore solvable, first-order difference equation.

2. Show, using the per capita production function, that $z_t = (k_t/A_t)^{1-\alpha}$. Then show, by writing this equation for period $t + 1$, using the definition $k_{t+1} \equiv K_{t+1}/L_{t+1}$, and then using the model's equations, that:

$$z_{t+1} = \left(\frac{1}{(1+n)(1+g)} \right)^{1-\alpha} (sy_t + (1-\delta)k_t)^{1-\alpha} \left(\frac{1}{A_t} \right)^{1-\alpha}.$$

From this equation show by appropriate manipulations that the law of motion for z_t is:

$$z_{t+1} = \left(\frac{1}{(1+n)(1+g)} \right)^{1-\alpha} z_t^\alpha (s + (1-\delta)z_t)^{1-\alpha}. \quad (51)$$

(Hint: Start by taking k_t out of the middle parentheses and then utilize the fact that inside the parentheses you get $y_t/k_t = 1/z_t$, and outside you get a $k_t^{1-\alpha}$ that you can put together with the $(1/A_t)^{1-\alpha}$ to make z_t appear.)

Clearly, (51) is a non-linear difference equation, so we did not get solvability.

3. Now derive the steady state value z^* directly from (51). Sketch the transition diagram for z_t , and establish properties of the transition equation that imply global, monotone convergence to z^* , from any given initial $z_0 > 0$. (For this you must establish properties as listed in Footnote 8 in Chapter 3.) Finally show that in steady state k_t and y_t must be increasing at the same rate, and that the common approximate growth rate must be g .

Exercise 3. Time to halve

In Section 5.4 we arrived at the dynamic equation, $\ln \tilde{y}_{t+1} - \ln \tilde{y}_t = \lambda(\ln \tilde{y}^* - \ln \tilde{y}_t)$, which is (34) with the rate of convergence λ inserted for $1 - G'(\tilde{k}^*)$, and equivalent to (36). Show that

it follows from this dynamic equation that the number of years, h , before half of an initial relative gap in year zero, $\ln \bar{y}^* - \ln \bar{y}_0$, has been covered is $h = -\ln 2 / \ln(1 - \lambda)$, and this is independent of how large the initial gap is. (Hint: Use the solution (37) to our difference equation.) For realistic parameter values and the expression (35) for λ , what size is h ? How large is h for our direct estimates of λ ?

Exercise 4. The effects of an increase in the savings rate and convergence time

In this chapter an increase in the savings rate was analysed in the Solow diagram and in the modified Solow diagram, and the growth jump arising from the increase in s was described qualitatively. It follows from the chapter that the speed by which the economy adjusts from the old to the new steady state should (approximately) be measured by the rate of convergence, $\lambda \approx (1 - \alpha)(n + g + \delta)$. This exercise asks you to do simulations and graphical representations to illustrate *numerically* the qualitative findings.

Let the parameters (before the shift) in the general Solow model be given by: $\alpha = \frac{1}{3}$, $s = 0.12$, $n = 0.01$, $g = 0.01$, $\delta = 0.05$. In the following, the *base scenario* is defined by the economy being in steady state at these parameter values from period zero through period 100.

1. Compute the steady state values $\bar{k}^*$ and $\bar{y}^*$ for $\bar{k}_t$ and $\bar{y}_t$ in the base scenario (all definitions are standard and as in the chapter). Illustrate the base scenario in three diagrams. Diagram 1 should plot $\bar{y}_t$ against time t (the points are just situated on a horizontal line). Diagram 2 should plot $\ln y_t$ against t , and Diagram 3 should plot $\ln c_t$ against t . You have not yet been given sufficient information to do Diagrams 2 and 3. Explain why. Let $A_0 = 1$, and do Diagrams 2 and 3.

The *alternative scenario* is defined by the economy evolving according to the base scenario from period zero through period nine (that is, all parameters, and A_0 , are as in the base scenario from period zero to period nine, and the economy is in steady state at these parameter values). In period ten a permanent shift in the savings rate from $s = 0.12$ to $s' = 0.24$ occurs, working initially in period ten, while all other parameters remain unchanged.

2. Simulate the model using appropriate parameters and initial values, thus creating computed time series for relevant variables according to the alternative scenario. Extend Diagram 1 so that it shows the evolution in $\bar{y}_t$ both according to the base scenario and according to the alternative scenario. Do the parallel extensions of Diagrams 2 and 3. In Diagrams 2 and 3 you should include the lines corresponding to the new steady state, that is, lines for $\ln y_t$ and $\ln c_t$ as if the economy had been in the new steady state in all periods. Now construct a Diagram 4 that shows the evolution in the growth rate, $g_t' \equiv \ln y_t - \ln y_{t-1}$, over the periods $t = 1, \dots, 100$, both according to the base and according to the alternative scenarios, and construct a Diagram 5 that shows the evolution in the capital–output ratio, K_t/Y_t (same periods, both scenarios).
3. From your simulations find the (approximate and rounded up) number of periods it takes for $\bar{y}_t$ to move half the way from its old to its new steady state value. In Exercise 3 you were asked to derive a formula for the half-life, h , given a rate of convergence, λ . Relate the half-life you have just found to the one computed from the formula of that exercise, using $\lambda \approx (1 - \alpha)(n + g + \delta)$. Why can there be a (slight) difference between these half-lives?
4. Do all of the above once more, but now with $g = 0.04$ throughout. So, the base scenario is now given by $\alpha = \frac{1}{3}$, $s = 0.12$, $n = 0.01$, $g = 0.04$, $\delta = 0.05$ and $A_0 = 1$, with the economy being in steady state at these parameter values in all periods, while the alternative scenario is given by an increase in the savings rate to 0.24 as from period ten, etc. Compare the two experiments in particular with respect to the half-lives of the adjustment from the old to the new steady state and relate to the ‘theoretical’ rate of convergence, λ .

Exercise 5. The effects of an increase in the rate of technological progress

Assume that the economy is initially in steady state at parameter values α, s, n, g, δ . Then from some period the rate of technological progress increases permanently to a new and higher constant level, $g' > 0$, but n, g and g' are relatively small, so $1/[(1+n)(1+g')]$ is approximately one, and ng' is approximately zero. This exercise asks you to conduct a qualitative analysis of the growth effects of the increase in the rate of technological progress. First, how are the Solow diagram and the modified Solow diagram affected by the change? In diagrams with t along the horizontal axis, illustrate how $\tilde{k}_t$ and the growth rate g_t^k of $\tilde{k}_t$ adjust over time (all definitions as usual). You should find that in the first period after the change, g_t^k falls to a certain negative value and then it increases monotonically back towards zero. Show that the value it first falls to is approximately $-(g' - g)$. Next, illustrate in a diagram how the approximate growth rate g_t^k of k_t evolves. In doing this, show that just after the change, g_t^k stays close to the original g , and then increases monotonically towards g' (here the formula $g_t^k = g_t^{\tilde{k}} + g_t^A$, following from $\tilde{k}_t = k_t/A_t$, is useful). Finally, show that the approximate growth rate, g_t^y , just after the change jumps from g to $\alpha g + (1 - \alpha)g'$, and from there increases monotonically towards g' (here $g_t^y = \alpha g_t^k + (1 - \alpha)g_t^A$ may be useful). Illustrate the adjustments of y_t in a $(t, \ln y_t)$ diagram.

Exercise 6. Deriving an estimate of α from the estimation of the convergence equation

In this chapter we derived an estimate of α from our estimation of the *steady state regression equation* (29). The γ in that equation was estimated to be 1.52 (see the displayed formula just after Fig. 5.7), and since $\gamma = \alpha/(1 - \alpha)$ according to the model, the value for α implied by the estimation had to be 0.6. Since α is capital's share of GDP, it should be around $\frac{1}{3}$, so we had to conclude that in this respect the model did not accord perfectly well with the data. From our estimation of the *convergence regression equation* (40) reported in (41) we derived an estimate of the rate of convergence, λ , and found a value of 0.7 per cent, which was low compared to the theoretical expression, $\lambda = (1 - \alpha)(n + g + \delta)$, since this is realistically around 5 per cent. We could also have derived an estimated value of α from our estimation of the convergence equation by using the estimated values of β_1 and β_2 and the theoretical equation (39). Find this implied value of α and compare it to the 'correct' value of around $\frac{1}{3}$, and to the value implied by the estimation of the steady state equation.

Exercise 7. Growth accounting: does your country keep up?

Table 5.2 shows GDP per worker, y_t , and non-residential capital per worker, k_t , for some countries and years. Use formula (45), assuming $\alpha = \frac{1}{3}$, to decompose for each country the average annual growth rate of GDP per worker over the full period 1966 to 1990 into components from growth in capital per worker and from TFP growth. Comment on how your own country has been doing (if your country is not there, just pick one as your home country for this exercise).

Next, use the same formula to do growth accounting for the sub-periods 1966–78 and 1978–90 for each country. Comment again with respect to how your country has been doing.

Exercise 8. Growth accounting: have the growth miracle economies experienced above-normal TFP growth?

Table 5.3 shows GDP per worker, y_t , capital per worker, k_t , and average years of education in the population aged 15 or more, for some East Asian and some Western countries for 1961 and 1990.

We all know that the East Asian countries included in the table have had very high growth in GDP per person compared to the West over the period considered, but have they also had relatively high technological growth? We argued in Section 5.5 that to give an answer via growth accounting, one would need to include education because some of the

TABLE 5.2 GDP per worker and non-residential capital per worker, 1985 – international prices

Year	USA		Denmark		UK	
	y_t	k_t	y_t	k_t	y_t	k_t
1966	29,152	18,734	18,124	15,284	16,895	9,362
1978	32,773	26,549	21,459	26,824	21,341	15,811
1990	36,771	34,705	24,971	33,125	26,755	21,179

Year	Ireland		Netherlands		Belgium	
	y_t	k_t	y_t	k_t	y_t	k_t
1966	10,374	7,538	20,958	15,674	18,267	16,330
1978	17,849	15,604	29,142	27,513	26,474	29,792
1990	24,058	21,660	31,242	32,390	31,730	36,646

Year	Norway		Sweden		Finland	
	y_t	k_t	y_t	k_t	y_t	k_t
1966	17,793	39,982	20,935	16,784	14,056	18,137
1978	23,984	41,875	23,935	28,173	19,244	32,496
1990	29,248	48,135	28,389	39,409	27,350	45,767

Note: This table is based on the older version 5.6 of Penn World Table. Some figures are substantially changed in later versions.

Source: Penn World Tables 5.6

TABLE 5.3 GDP per worker, capital per worker, and years of education, East and West, 1961 and 1990

East	Japan			Hong Kong		
	y_t	k_t	u_t	y_t	k_t	u_t
1961	5,630	7,601	7.8	4,372	7,294	5.2
1990	22,624	85,741	9.0	22,827	30,553	9.2

Year	South Korea			Singapore		
	y_t	k_t	u_t	y_t	k_t	u_t
1961	2,724	2,034	4.2	5,225	8,445	4.3
1990	16,022	28,393	9.9	24,369	56,670	6.0

West	USA			UK		
	y_t	k_t	u_t	y_t	k_t	u_t
1961	24,431	52,113	8.5	15,077	21,214	7.8
1990	36,771	80,419	11.7	26,755	49,227	8.8

Year	France			Italy		
	y_t	k_t	u_t	y_t	k_t	u_t
1961	14,111	26,467	5.4	11,987	27,560	4.7
1990	30,357	80,194	7.0	30,797	78,026	6.5

Year	Netherlands			Denmark		
	y_t	k_t	u_t	y_t	k_t	u_t
1961	17,501	35,747	5.3	15,554	28,599	9.0
1990	31,242	72,234	8.7	24,971	64,480	9.6

Note: u_t is average years of education of people older than 15 years. y_t and k_t are measured in 1985 – international prices.

Sources: William Easterly and Ross Levine, 'It's Not Factor Accumulation: Stylized Facts and Growth Models', *The World Bank Economic Review* 15, No.2, 2001; Robert J. Barro and Jong-Wha Lee, *International Data on Educational Attainment*, Center for International Development at Harvard University (CID), 2000.

East Asian countries have had large increases in education standards, as also evidenced by the numbers in the table. This exercise asks you to do some growth accounting with education included. Let the aggregate production function for each country be:

$$Y_t = K_t^\alpha (A_t h(u_t) L_t)^{1-\alpha}, \quad h(u_t) = \exp(\psi u_t),$$

where we have chosen this time to express technology in terms of the labour augmenting productivity variable, A_t .

1. Derive an appropriate growth accounting formula that splits up the average annual growth rate of GDP per worker into components coming from growth in A_t , growth in capital per worker, and growth in education.
2. Use the formula to split up the average annual growth rates of GDP per worker from 1961 to 1990 into the components described in Question 1 for the countries shown in the table, assuming $\alpha = \frac{1}{3}$, $\psi = 0.1$.
3. Discuss whether there seems to have been excessive technological progress in East Asia. Can this growth accounting exercise, by reasonable standards, be said to give a full answer to the question raised?

Exercise 9. Where growth comes from according to the Solow model and according to growth accounting

Assume that some economy behaves completely in accordance with the steady state of the Solow model presented in this chapter, where the assumed growth rate of A_t is g . From what you have learned in this chapter, what will the average annual growth rates of y_t , k_t and A_t be over some period from year t to year T ? What creates growth in GDP per worker? If one did a growth accounting exercise for this economy, what would be the fractions of growth in GDP per worker attributed to growth in capital per worker and to growth in A_t , respectively? Again, what is the true source of growth in this economy?

Exercise 10. One (more) critical look at the empirical success of the Solow model's steady state prediction

In this chapter we concluded, based on Fig. 5.7 and the estimation of (29) going with it, that the steady state prediction of the Solow model did *quite* well empirically (but with some shortcomings concerning magnitudes of estimates). One thing to learn about empirical analysis is that there is always reason to be critical and to ask hard questions. This exercise illustrates one standard type of critical question: is the conclusion arrived at robust with respect to an appropriate division of the available data? All required data can be taken from Table A at the back of Book One. Proceed as follows.

Consider the countries in the sample of 65 countries. Split these countries into two subgroups: countries that in 2003 had a GDP per worker above 50 per cent of the US level, and countries that had GDP per worker below 50 per cent of the US level. For each of these groups, construct a figure like Fig. 5.7, and estimate by OLS the steady state equation (29) for each subgroup, that is, find the lines of best fit in your figures. Comment on your findings.

(Exercise 1 of Chapter 6 will illustrate another standard type of critical question one can ask, still in connection with the empirical success of the steady state prediction of the Solow model. If you wish you can, for your own interest, already do most of that exercise now, but it has a close relationship to the analysis in Chapter 6 as well.)